A nonlocal game whose dimension exponent is uncomputable

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Version 1.0.0, October 2026

Abstract

Can uncomputability occur in the resource law of one fixed finite nonlocal game, rather than in a problem whose input is a game? We construct a finite two-player Boolean constraint-system game with rational question probabilities, quantum value one, and no finite-dimensional perfect strategy. If $D(\varepsilon)$ is the least local dimension needed to win with probability at least $1-\varepsilon$, then

$$\lim_{n\to\infty} \frac{\log_2 D(2^{-n})}{n} = \frac{3+\theta_K}{4}, \qquad \theta_K = \sum_{e\in K} 4^{-(e+1)},$$

where K is the halting set. This exponent has the Turing degree of K. The construction combines finite operator relations for three Cantor clocks, a counter program with forced initialization, and a conditional clock that charges dimension for halting inputs. A Boolean compiler preserves weighted precision linearly and converts flat models to strategies with quadratic loss. The lower bound holds in every density matrix. Explicit finite models attain the same rate. The flag-density lower bound uses compactness without an effective modulus; convergence to the dimension exponent has no computable certified rate.

The authors developed this work with current frontier AI models from more than one developer and take responsibility for every claim. How the work was done and checked, including a refutation-first proof audit and finite computations, is described before the appendices.

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1 One fixed game

For a finite nonlocal game, near-perfect play can require arbitrarily large local dimension. We ask whether the exponent of that growth can be uncomputable for one fixed game. This places the uncomputable real inside a single finite specification: the input to the asymptotic problem is only the desired accuracy. Undecidability of a decision problem in which the game varies does not answer this question.

Fix a standard effective enumeration of programs and let $K \subseteq \mathbb{N} = \{0, 1, \ldots\}$ be its halting set. All logarithms without a displayed base are base two. For a game G of quantum value one, let $D_G(\varepsilon)$ be the least integer d for which a strategy on $\mathbb{C}^d \otimes \mathbb{C}^d$ wins with probability at least $1 - \varepsilon$. Smaller local spaces may be padded.

Theorem 1.1. *There is a fixed finite two-player Boolean constraint-system game G, with rational question probabilities, such that $\omega_q(G) = 1$, no finite-dimensional strategy wins perfectly, and*

$$D_G(\varepsilon) = \varepsilon^{-(3+\theta_K)/4+o(1)}, \qquad \theta_K = \sum_{e \in K} 4^{-(e+1)}. \tag{1.1}$$

The exponent is Turing-equivalent to K. The sequence $\log D_G(2^{-n})/n$ has no computable certified rate of convergence.

The theorem concerns local dimension, not distillable entanglement. Its game is an ordinary Boolean constraint-system (BCS) game, rather than a linear system game. The computable baseline of the exponent is $3/4$. These distinctions delimit what is proved; the corresponding distillation rate, a linear-system realization with this rate, and a smaller baseline remain open.

The existence of computable increasing rational sequences with uncomputable limits is classical, beginning with Specker [25]. The issue here is their realization as the dimension law of a finite game. The separated base-four encoding of K is used only to make its Turing degree explicit.

There are three main points. First, a finite set of bounded polynomial relations can be compiled into Boolean constraints without changing the asymptotic precision scale. Second, a finite program can be forced to start at every finite fuel scale, while finite models still close with exponentially small defects. Third, a dimension count on the actual density matrix converts a non-effective flag-density limit into a matched dimension exponent.

1.1 Related work and credit

Mousavi, Nezhadi and Yuen prove that deciding whether the quantum value is exactly one is Π^0_2-complete [16, 17]. In particular, there is no total computable dimension bound, uniform over all value-one games and all precisions. Indeed, such a bound would make value one a universal integer quantification of decidable bounded-dimensional feasibility statements, placing that problem in Π^0_1. Their obstruction is uniform over games. Theorem 1.1 instead identifies a convergent exponent for one game. The theorem MIP* = RE puts halting information in the value of a computable family of games [13]; it does not by itself identify a resource exponent of a fixed value-one game.

The algebraic connection between constraint games and operator assignments goes back to Cleve and Mittal [4] and, for linear systems, Cleve, Liu and Slofstra [3]. Slofstra established an embedding theorem for solution groups [23] and the non-closure of quantum correlations [22]. Slofstra and Vidick introduced the hyperlinear profile as a quantitative bridge from approximate representations to dimension requirements [24]. Slofstra later obtained a finitely presented group with at least subexponential hyperlinear profile [21]. Ji, Leung

and Vidick construct a fixed finite three-player game of value one for which near-perfect play requires at least $\Omega(\varepsilon^{-c})$ qubits for a fixed $c > 0$ [12]. Coladangelo constructs a fixed finite two-player score game for which every ε-optimal strategy has local dimension $2^{\Omega(\varepsilon^{-1/8})}$ [5]. These results supply the framework and important growth examples; they do not identify the matched finite exponent in (1.1).

Mehta, Slofstra and Zhao construct halting-dependent Pauli systems and embed them in finite presentations with polynomial bounds on relation decompositions [14, Proposition 4.8]. Those bounds transfer lower estimates; they do not assert small simultaneous finite-model extensions. The construction here supplies its own upper models. The conjugacy gadgets used in our compiler are those of Fanizza, Kroell, Mehta, Paddock, Rochette, Slofstra and Zhao [8, Lemma 5.8 and Theorem 5.5]. Their credit is essential: the passage from nested conjugacy to ordinary Boolean contexts is not a new gadget of this paper. We give its matrices and converse calculation because quantitative completeness is needed here.

The state-preserving Boolean correspondence was established in *The price of precision* [7, Theorem 4.9]; its linear-system predecessor and further dimension examples appear in the companion paper [15]. We use that correspondence as an existing theorem. Section 10 records the elementary square-root estimate needed for the present local-dimension application and checks the constants' effect on the exponent. The construction partially addresses the rate question of [7, Problem 18.3], for a different resource: local dimension.

Uncomputable asymptotic data in finite systems have other precedents. Hochman and Meyerovitch characterize the entropies of multidimensional shifts of finite type and realize noncomputable extremal frequencies [10]. Sapir, Birget and Rips relate group isoperimetry and computation [19]; a Dehn exponent is not a dimension exponent, and no identification between them is used here. Cubitt, Perez-Garcia and Wolf prove undecidability of the spectral gap over families of local Hamiltonians [6]. Bausch, Cubitt and Watson obtain uncomputable phase diagrams for a continuous one-parameter family [1]. These are precedents for asymptotic noncomputability, rather than proofs of (1.1).

The binary adding machine used below is the classical odometer on a Cantor set; its interpretation through operator algebras is standard [9]. The new point needed here is a finite polynomial presentation and a carry estimate that remain quantitative in a non-flat state. The final information bound uses Holevo's theorem [11] and the random-access-code principle exemplified by Nayak [18]. Its branching estimate is proved explicitly.

1.2 Proof outline

At precision 4^{-N}, three compulsory clocks cost $3N - o(N)$ bits of logarithmic dimension. They hold an address, fuel, and time. A finite counter program recognizes the halting set. Reset relations force an ordinary initialized computation on each fixed address cylinder and each sufficiently long fuel sector. Exact tracial models must therefore flag every halting address. Compactness extends the flag-density conclusion uniformly to approximate models, without an effective modulus.

A fourth, conditional clock contributes one further bit at each active address. Its contribution is $\theta_K N - o(N)$. In the upper models, the program is a collection of diagonal labels on the existing clock registers. Its histories require no further tensor factor. A moving register of optional bits realizes the conditional clock in dimension $2^{\theta_N N + O(\log N)}$, where $\theta_N \uparrow \theta_K$ is computable. Source precision 4^{-N} becomes game loss of order 16^{-N}. This accounts for the denominator four in (1.1).

2 Weighted models and elementary estimates

Definition 2.1. A bounded polynomial presentation consists of finitely many variables, each declared to be a positive contraction, a projection, a self-adjoint unitary, or a unitary, and finitely many rational real-coefficient noncommutative $*$-polynomial equations $r = 0$. Its weighted precision-δ model on $\mathbb{C}^d$ is an assignment in the declared domains and a density matrix ρ such that, with $\xi = \sqrt{\rho}$,

$$\|r\xi\|_2 \le \delta, \qquad \|[s, \xi]\|_2 \le \delta \tag{2.1}$$

for each defining relation r and each variable s. The norm is the unnormalized Hilbert–Schmidt norm. A flat model has $\rho = I/d$ and $\|r\|_f = d^{-1/2}\|r\|_2$. Write $W(\delta)$ and $H(\delta)$ for the least weighted and flat model dimensions, with value ∞ if no model exists.

We write $|D|_\rho = \|D\xi\|_2$. Constants for a fixed presentation may depend on its finite polynomials, but never on d, ρ, δ or a growing digit depth.

Lemma 2.2 (Right multiplication)**.** *For bounded A, B,*

$$|AB|_\rho \le \|A\||B|_\rho, \qquad |BA|_\rho \le \|A\||B|_\rho + \|B\|\|[A, \xi]\|_2.$$

A word w of length l in contraction variables and unitary adjoints satisfies $\|[w, \xi]\|_2 \le l\delta$. A fixed finite algebraic derivation from the defining relations has error at most $C\delta$ on the same state.

Proof. Use $BA\xi = B\xi A + B[A, \xi]$. Expand the commutator of a product one factor at a time. Adjoint unitaries have the same state-commutator norm as the unitary. For adjoint relations,

$$|r^*|_\rho = \|\xi r\|_2 \le |r|_\rho + \|[r, \xi]\|_2.$$

All variables are bounded, so a fixed polynomial has a fixed operator-norm bound and a commutator bound $C\delta$. Expand a finite ideal identity as a finite sum $a_j r_j b_j$ and apply the first two inequalities. This proves the last assertion. □

Lemma 2.3 (Hilbert–Schmidt functional calculus)**.** *Let A, B be self-adjoint, Z an arbitrary rectangular Hilbert–Schmidt matrix, and g an L-Lipschitz function on their spectra. Then*

$$\|g(A)Z - Zg(B)\|_2 \le L\|AZ - ZB\|_2, \tag{2.2}$$

$$|g(A) - g(B)|_\rho \le L\big(|A - B|_\rho + 2\|[B, \xi]\|_2\big), \tag{2.3}$$

$$|[Y, g(A)]|_\rho \le L\big(|[Y, A]|_\rho + 2\|[A, \xi]\|_2\big) \tag{2.4}$$

in the last line for a contraction Y. Also $\|[g(A), \xi]\|_2 \le L\|[A, \xi]\|_2$.

Proof. Diagonalize A on the left and B on the right. Corresponding entries in the first inequality are $(g(a) - g(b))Z_{ab}$ and $(a - b)Z_{ab}$. Square the scalar Lipschitz inequality and sum. The state-commutator estimate is its case $A = B, Z = \xi$. Insert and subtract $\xi g(B)$ to obtain (2.3). For (2.4), apply (2.2) with $A = B, Z = Y\xi$; then insert and subtract $Y\xi g(A)$. The two resulting state-commutator terms are each bounded by $L\|[A, \xi]\|_2$. These are Schatten-two estimates, not operator-norm Lipschitz statements. □

Lemma 2.4 (Finite-precision computability)**.** *For a rational-data game promised to have quantum value one, $D_G(\varepsilon)$ is computable at every positive rational ε. If $a_n = \log D_G(2^{-n})/n$ converges to β and $|a_n - \beta| \le r(n)$ eventually for a computable $r(n) \to 0$, then β is computable.*

Proof. In a fixed local dimension d, introduce real variables for the real and imaginary matrix entries of the state and both families of POVMs. Positivity is a finite list of principal-minor inequalities. Normalization and winning probability at least $1-\varepsilon$ are polynomial conditions with rational coefficients. Real quantifier elimination decides feasibility. Test $d = 1, 2, \ldots$; the value-one promise ensures termination. For the second assertion, hardcode a threshold beyond which the error bound holds. Given k, search above it for n with $r(n) < 2^{-k-1}$ and compute a_n within 2^{-k-1}. This gives a 2^{-k} approximation to β. □

3 From bounded polynomials to Boolean constraints

A BCS specifies finitely many Boolean variables and finitely many contexts. A context lists variables and its allowed sign assignments. In an operator model the context variables are commuting reflections and their joint spectrum is restricted to those assignments. Its game asks Alice for a valid context assignment and Bob for one sign, testing consistency on an incidence. We use the uniform distribution on the finite incidence set, hence rational probabilities. The algebraic interpretation is standard [4, 3].

Theorem 3.1 (Polynomial compiler)**.** *For each finite bounded polynomial presentation $\mathcal{P}$, there are a finite ordinary BCS $\mathcal{B}$ and constants $c, C > 0$ with the following properties.*

(a) Every weighted precision-η model of $\mathcal{B}$ reconstructs a precision-$C\eta$ model of $\mathcal{P}$ on the same space and with the same state.

(b) Every flat precision-δ model of $\mathcal{P}$ on $\mathbb{C}^d$ gives a strategy for its context-variable game in local dimension at most cd, with loss at most $C\delta^2$.

All constants depend only on the presentation. In particular, guarded equations such as $(V - W)(I - A) = 0$ are compiled with their guard intact.

3.1 Domains and selectors

Every positive contraction is

$$r = \frac{2I + u + u^*}{4}, \qquad u = 2r - I + 2i\sqrt{r(I-r)}. \tag{3.1}$$

The real and imaginary parts in this expression commute and their squared sum is I, so u is unitary. Conversely the first formula is a positive contraction for any unitary. For upper models realify the whole assignment by

$$\mathcal{R}(M) = \begin{pmatrix} \Re M & -\Im M \\ \Im M & \Re M \end{pmatrix}.$$

This doubles dimension, preserves real-coefficient equations and flat norms, and takes unitaries to real orthogonal matrices. A real orthogonal matrix is a product of two symmetric orthogonal involutions: on a rotation plane use

$$\begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cos t & -\sin t \\ -\sin t & -\cos t \end{pmatrix},$$

and use scalar blocks on the remaining eigenspaces. Orthogonal conjugation and direct sums finish the factorization. Replace each unitary by a product of two reflections. Lower reconstructions are automatically unitary, and (3.1) is automatically positive. Their state

commutators cost a fixed factor. No commutation between the two factors is imposed. The upper models are flat, so no state estimate for the square root in (3.1) is needed.

A source equation is now $p = \sum_j a_j W_j = 0$, with rational a_j and reflection words W_j. Clear denominators with an integer c, duplicate signed terms, and add cancelling pairs until

$$\sum_{j=1}^{L} \widetilde{W}_j = 2cp, \qquad L = 2^q.$$

Introduce q Pauli pairs X_i, Z_i, commuting across pairs and with every source reflection. Let e_j be the joint Z spectral projections and E the joint $+1$ projection of the X_i. Then

$$Ee_jE = L^{-1}E, \qquad \sum_{v\in\{0,1\}^q} Z^v E Z^v = I. \tag{3.2}$$

For the first identity, expand e_j in Z monomials. Every nonconstant monomial anticommutes with some X_i and hence has zero compression by E. The second identity sums the joint X spectral projections.

The controlled reflection $(I - e_j) + e_j s$ is a Boolean truth table in commuting variables. Products of these controls and controlled signs form a reflection word whose value is the unitary

$$S = \sum_j e_j \widetilde{W}_j.$$

The projection E receives its own Boolean variable x_E. Its defining context is $\{X_1, \dots, X_q, x_E\}$, whose allowed assignments are those with $x_E = +1$ exactly when $X_1 = \cdots = X_q = +1$. In an operator model this truth table says $(I + x_E)/2 = \prod_i (I + X_i)/2 = E$. Introduce $x_F = S x_E S^*$ by successive reflection conjugacies and put $F = (I + x_F)/2 = SES^*$. Orthogonality is imposed on this pair alone: the context $\{x_E, x_F\}$ forbids exactly the assignment $(+1, +1)$, which in an operator model is the condition $EF = 0$. The exact identities are

$$EFS = ESE = \frac{2c}{L} pE, \qquad p = \frac{L}{2c} \sum_v Z^v EFS Z^v. \tag{3.3}$$

They show why a low-weight selector corner cannot conceal an equation. All of its translates are used.

3.2 The conjugacy gadgets

We use the gadgets of [8, Lemma 5.8]. Their formulas are included to establish the precise finite-dimensional extension needed here. For a general relation $x_1 x_2 x_1 = x_3$, introduce commuting parity contexts

$$x_i y_i z_i = I, \qquad x_i f w_i = I, \qquad g y_2 z_3 = I,$$

and the conjugacies $f y_i f = z_i$ and $w_1 y_2 w_1 = z_3$. The targets of these remaining conjugacies commute by the parity contexts. If the original reflections are X_i, an exact doubled extension is

$$x_i = \operatorname{diag}(X_i, X_i), \quad y_i = \operatorname{diag}(X_i, I), \quad z_i = \operatorname{diag}(I, X_i),$$
$$f = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}, \quad w_i = \begin{pmatrix} 0 & X_i \\ X_i & 0 \end{pmatrix}, \quad g = \operatorname{diag}(X_2, X_3).$$

Conversely $w_1 = x_1 f$, $[f, x_1] = 0$, and $f y_2 f = z_2$ give $x_1 z_2 x_1 = z_3$. Conjugation by f gives $x_1 y_2 x_1 = y_3$. Multiplying gives the required conjugacy.

For $aba = c$ with commuting targets b, c, use six commuting parity contexts:

$$ad_1d_2 = bd_2d_3 = d_3d_4d_5 = ad_5d_6 = cd_6d_7 = d_1d_4d_7 = I. \tag{3.4}$$

They imply

$$aba = d_1d_2d_2d_3d_5d_6 = d_1d_4d_6 = d_7d_6 = c.$$

Writing $M(a) = \left(\begin{smallmatrix} 0 & a \\ a & 0 \end{smallmatrix}\right)$, an exact extension, with old variables repeated diagonally, is

$$d_1 = M(a), \quad d_2 = M(I), \quad d_3 = M(b), \quad d_4 = \begin{pmatrix} 0 & ba \\ ab & 0 \end{pmatrix},$$
$$d_5 = \mathrm{diag}(bab, a), \quad d_6 = \mathrm{diag}(bc, I), \quad d_7 = \mathrm{diag}(b, c).$$

For clarity, $d_4^* = d_4$ and $d_4^2 = I$. The equality $aba = c$ implies that a commutes with bc, while bc is a reflection. These facts verify the fourth context; the other five follow by direct multiplication. Each listed triple consists of commuting reflections with product I. Sequentially applying the two gadgets has constant dimension cost for a fixed presentation. Pauli anticommutation is $XZX = -Z$, with a Boolean variable for the negation of Z.

3.3 Weighted soundness and flat completeness

Proof of Theorem 3.1. All identities above are finite algebraic calculations in the defining relations. A Boolean truth table gives finite algebraic relations: expand in its assignment projections, and any polynomial vanishing on allowed assignments is the finite sum of its values on forbidden assignments times their projections. Thus the controlled-variable definitions, both gadget converses and (3.3) are finite ideal derivations. In an approximate model put $E_{\mathrm{pol}} = \prod_i (I + X_i)/2$, $E_0 = (I + x_E)/2$ and $F_0 = (I + x_F)/2$. The defining AND context of x_E and the conjugacy gadgets give $|E_{\mathrm{pol}} - E_0|_\rho = O(\eta)$ and $|SE_{\mathrm{pol}}S^* - F_0|_\rho = O(\eta)$, while the orthogonality context gives $|E_0F_0|_\rho = O(\eta)$. Interpret (3.3) as a polynomial identity modulo the Pauli and controlled-variable relations. Substitution of these estimates, using Lemma 2.2, gives $|p|_\rho = O(\eta)$. Only E_0 and F_0 are automatically projections here; the selector polynomial need not be a projection in an approximate model. Lemma 2.2 proves linear weighted soundness. This statement is stronger than mere exact implication; the explicit finite derivations are what justify the linear error.

For upper models choose the selectors on an exact tensor factor and define every controlled variable and successive conjugate projection canonically. Put $x_E = 2E - I$ with $E = \prod_i (I + X_i)/2$. It commutes with every X_i, and the joint spectrum of $(X_1, \dots, X_q, x_E)$ lies in the allowed assignments, so the defining context of x_E is exact. The conjugacies defining x_F are exact even when p is nonzero. All gadget contexts extend exactly. The only inexact constraint is the final orthogonality, where

$$\|EF\|_f = \|ESE\|_f = \frac{2c}{L\sqrt{L}}\|p\|_f. \tag{3.5}$$

Let F' be the projection onto $(I - E)\operatorname{ran} F$. Then

$$EF' = 0, \qquad \|F - F'\|_f^2 \le 2\|EF\|_f^2. \tag{3.6}$$

To prove this, diagonalize FEF on $\operatorname{ran} F$ with eigenvalues λ_i and orthonormal eigenvectors v_i. For $\lambda_i < 1$, the vectors $(I - E)v_i/\sqrt{1 - \lambda_i}$ are orthonormal and span the new range. Those with $\lambda_i = 1$ disappear. Hence

$$\|F - F'\|_2^2 = \#\{i : \lambda_i = 1\} + 2\sum_{\lambda_i < 1} \lambda_i \le 2\sum_i \lambda_i = 2\|EF\|_2^2.$$

On the orthogonality context $\{x_E, x_F\}$, Alice answers with the PVM

$$E \mapsto (+1,-1), \qquad F' \mapsto (-1,+1), \qquad I-E-F' \mapsto (-1,-1).$$

These are orthogonal projections summing to I, because $F' \le I-E$, and every outcome is an allowed assignment. Her operator for x_E is therefore $2E-I$ and her operator for x_F is $2F'-I$. Bob uses $2E-I$ and $2F-I$. On a maximally entangled state there is no disagreement for x_E, and the disagreement probability for x_F is $\|F-F'\|_f^2$. This is why E carries its own variable: if the context listed $X_1,\dots,X_q$ together with x_F, the PVM above would not be a legal answer, since F' commutes with E but in general not with each X_i, and so does not fix their values. All other contexts, including the defining context of x_E, are exact. With incidence probability π, summing over the finitely many source equations and using (3.6) and (3.5) bounds the loss by

$$2\pi \sum \|EF\|_f^2 = 8\pi \sum \frac{c^2}{L^3}\|p\|_f^2 \le C\delta^2.$$

Here c and L are the integers of each source equation. The new variable adds no tensor factor. Realification, selectors and gadget extensions multiply the dimension by the fixed factor $2^{1+q_{\mathrm{tot}}+g}$, where q_{tot} is the total number of selector qubits and g the number of gadget doublings; this is the dimension constant in (b). This proves both assertions. □

4 A finite Cantor clock

The clock has a positive contraction A, a projection P, a unitary T and a self-adjoint unitary X, with relations

$$[A,P]=0, \quad TAT^* = 4A-3P, \quad XPX = I-P, \quad [X,TAT^*]=0. \tag{4.1}$$

For $j \in \mathbb{N}$ put $P_j = T^j P T^{-j}$ and $X_j = T^j X T^{-j}$. The Cantor set used here is

$$\mathcal{K} = \left\{ \sum_{j\ge 0} 3b_j 4^{-j-1} : b_j \in \{0,1\} \right\}.$$

Let p be its first-digit function, extended to $[0,1]$ by

$$p(a) = \begin{cases} 0 & a \le 1/4, \\ 2a-1/2 & 1/4 \le a \le 3/4, \\ 1 & a \ge 3/4, \end{cases} \qquad s(a) = 4a - 3p(a).$$

These functions are respectively 2- and 4-Lipschitz and map $[0,1]$ into itself.

Lemma 4.1 (Exact digits)**.** *In an exact model, A has spectrum in $\mathcal{K}$, all P_j are commuting spectral functions of A, and*

$$A = \sum_{j<M} 3\cdot 4^{-j-1} P_j + 4^{-M} T^M A T^{-M}. \tag{4.2}$$

The flip X_j exchanges P_j and $I-P_j$ and preserves all P_l with $l>j$. In every tracial state,

$$\tau\left(\prod_{j<M} P_j^{b_j} (I-P_j)^{1-b_j} \right) = 2^{-M}. \tag{4.3}$$

The spectral projection of A at 1 is zero in the faithful tracial representation.

Proof. Since A, P commute and $0 \le 4A - 3P \le I$, the $I - P$ block of A has spectrum in $[0, 1/4]$ and the P block in $[3/4, 1]$. Hence $P = p(A)$ and $TAT^* = s(A)$. Conjugating this statement repeatedly proves $P_j = p(s^j(A))$ and (4.2). Its remainder has norm at most 4^{-M}, which places the spectrum in $\mathcal{K}$. The tail coordinate TAT^* determines every later digit. Its commutation with X, and $XPX = I - P$, give the flip assertions, then their translates. A trace assigns equal mass to a cylinder and its earliest-bit flip. Add the two and repeat on the remaining suffix to obtain (4.3). The spectral projection at 1 lies under every all-ones cylinder and has trace at most 2^{-M} for every M. Faithfulness makes it zero. □

Lemma 4.2 (Linear spectral repair)**.** *There is an absolute constant C such that any weighted precision-δ clock has a self-adjoint $\widehat{A}$ with spectrum in $\mathcal{K}$ satisfying*

$$|\widehat{A} - A|_\rho + \|[\widehat{A}, \xi]\|_2 \le C\delta, \tag{4.4}$$

$$|P_j - p(s^j(\widehat{A}))|_\rho \le C4^j\delta. \tag{4.5}$$

The first M raw digit projections and triangular flips satisfy digit–digit commutations, own-bit flips, later-digit preservation and state commutations with error at most $C(M+1)^3 4^M\delta$. The spectral digit projections $p(s^j(\widehat{A}))$ commute exactly and differ from the raw ones by (4.5).

Proof. Set $B_0 = TAT^*$, $D_0 = PB_0P + (I-P)B_0(I-P)$ and $C_0 = (D_0 + 3P)/4$. The affine relation and $[A, P]$ give $|[B_0, P]|_\rho = O(\delta)$ by Lemma 2.2. Since $B_0 - D_0 = (2P - I)[P, B_0]$, it follows that $|D_0 - B_0|_\rho = O(\delta)$ and $|C_0 - A|_\rho = O(\delta)$. Their state commutators are $O(\delta)$ from their fixed polynomial definitions. All are positive contractions, and D_0 commutes with P.

Let $d_\mathcal{K}$ be distance to $\mathcal{K}$, a 1-Lipschitz function. Cantor self-similarity on the two P blocks gives $d_\mathcal{K}(C_0) = d_\mathcal{K}(D_0)/4$. Lemma 2.3 and unitary transport give

$$|d_\mathcal{K}(A)|_\rho \le \tfrac{1}{4}|d_\mathcal{K}(A)|_\rho + O(\delta).$$

Thus $|d_\mathcal{K}(A)|_\rho = O(\delta)$. Round each spectral value of A to a nearest point of $\mathcal{K}$. The resulting $\widehat{A}$ has $|\widehat{A} - A|_\rho = O(\delta)$. For any self-adjoint E, $\|[E, \xi]\|_2 \le 2|E|_\rho$; apply this to $E = \widehat{A} - A$ to prove (4.4). No continuity of the nearest-point map is needed.

Since $p(C_0) = P$, functional calculus gives $|P - p(\widehat{A})|_\rho = O(\delta)$, and then the affine relation gives $|T\widehat{A}T^* - s(\widehat{A})|_\rho = O(\delta)$. For $g = p \circ s^r$, the one-step error

$$|Tg(\widehat{A})T^* - g(s(\widehat{A}))|_\rho \le C4^r\delta$$

follows from (2.3). Telescope these identities to depth j. The sum is $O(4^j\delta)$. Every digit function has norm at most one, so transporting the errors through the outer powers of T adds only $O(j^2\delta)$, absorbed by $O(4^j\delta)$. This proves (4.5).

For the flips, translate $XPX = I - P$ and replace each P_j using (4.5). For $l > j$, use the tail coordinate $T^{j+1}AT^{-(j+1)}$. The translated last relation of (4.1) makes its commutator with X_j at most $C(j+1)\delta$ in weighted norm. Its digit at position $l-j-1$ is a function with Lipschitz constant at most $2 \cdot 4^{l-j-1}$. Apply (2.4), replace the digit back, and use Lemma 2.2. The error is at most $C(M+1)^3 4^M\delta$. Raw digit commutators follow by replacing both with commuting functions of $\widehat{A}$; the two replacement costs add. Finally $\|[P_j, \xi]\|_2, \|[X_j, \xi]\|_2 \le (2j+1)\delta$. This proves all assertions. □

Remark 4.3. The spectral repair and digit estimate use only positivity, the affine equation, $[A, P] = 0$, and state commutators. They apply without a flip. This will be used for the conditional coordinates.

Theorem 4.4 (Clock profile)**.** *For the presentation* (4.1)*,*

$$\log W(2^{-n}) = \frac{n}{2} + O(\log(n+2)), \qquad \log H(2^{-n}) = \frac{n}{2} + O(\log(n+2)).$$

There are flat models on N qubits of precision $3 \cdot 4^{-N}$.

Proof. On N qubits let P_j read bit j, let T rotate contents from slot j to slot $j+1$ cyclically, and let X flip bit zero. Put

$$A_N = \sum_{j=0}^{N-1} 3 \cdot 4^{-j-1} P_j. \tag{4.6}$$

Then $[A_N, P_0] = 0$, $XP_0X = I - P_0$, and

$$TA_NT^* - 4A_N + 3P_0 = 3 \cdot 4^{-N} P_0, \qquad [X, TA_NT^*] = 3 \cdot 4^{-N}[X, P_0].$$

All state commutators vanish in the flat state, proving the upper bound. For the lower bound, use M digits and flips from Lemma 4.2 in the binary case of the counting lemma, Theorem 8.1, proved below. Choose $M = \lfloor \log_4(1/\delta) - 20 \log_4(2 + \log_4(1/\delta)) \rfloor$. Its error terms tend to zero, so $\log d \geq M - o(1)$. At $\delta = 2^{-n}$ this is $n/2 - O(\log(n+2))$. Since $W \leq H$, both conclusions follow. This forward reference uses no property of the program constructed later. □

4.1 Guarded increment

Add a unitary V and the relation

$$(V - X[(I - P) + (TVT^*)P])(I - A) = 0. \tag{4.7}$$

Proposition 4.5. *The finite clocks have increment models of dimension 2^N whose guarded defect has operator norm $\sqrt{2}\, 4^{-N}$ and flat norm $\sqrt{2}\, 4^{-N} 2^{-N/2}$. In an exact faithful tracial model, V increments every finite binary digit prefix.*

Proof. Let V_N add one modulo 2^N, with bit zero least significant. If that bit is zero, the right-hand side before the guard is the correct first-bit flip. If it is one and some later bit is zero, TV_NT^* carries through slots $1, 2, \ldots$ to the first zero and X then clears slot zero. Only the all-ones input fails:

$$R_N := V_N - X[(I - P_0) + (TV_NT^*)P_0] = (|0^N\rangle - |10^{N-1}\rangle)\langle 1^N|.$$

The guard equals 4^{-N} there. The two norm statements follow from the rank-one formula. For exact tracial models, Lemma 4.1 makes $I - A$ have dense range, so its bounded left factor in (4.7) vanishes. Put $C = TVT^*$ and $D = (I - P) + CP$. Since $V = XD$, D is unitary. It fixes $\ker P$ pointwise, so unitarity forces $\operatorname{ran} P$ onto itself; there D is C. Hence $[C, P] = 0$. The recursion now reads binary increment on the first bit followed by increment on the tail on its carry branch. Induction on prefix length proves the assertion. □

Proposition 4.6 (Mass near the vanishing guard)**.** *There is an absolute constant C such that every weighted precision-δ clock and every integer $M \geq 1$ satisfy*

$$\operatorname{Tr}(\rho\, 1_{[1-4^{-M},1]}(A)) \leq 2^{-M} + C(M+1)^5 4^M \delta. \tag{4.8}$$

If $\|R\| \leq L$ and $|R(I - A)|_\rho \leq \delta$, then

$$|R|_\rho \leq 4^M(1 + L)\delta + L\sqrt{2^{-M} + C(M+1)^5 4^M \delta}. \tag{4.9}$$

In particular $|R|_\rho = O((1 + \log(1/\delta))^{5/2} \delta^{1/6})$ for fixed L.

Proof. Here it is useful to avoid rounding A first. From the proof of Lemma 4.2, $|P-p(A)|_\rho = O(\delta)$ and $|TAT^* - s(A)|_\rho = O(\delta)$. The same telescoping gives $|P_j - p(s^j(A))|_\rho \le C4^j\delta$. Threshold $p(s^j(A))$ at $1/2$ to commuting projections R_j. Against an exact projection the cross-spectrum Lipschitz bound for this threshold is two: for $b \in \{0,1\}$ and $a \in [0,1]$, $|1_{[1/2,1]}(a)-b| \le 2|a-b|$. The eigenbasis proof of Lemma 2.3, followed by its state-commutator insertion, therefore gives $|R_j-P_j|_\rho \le C(j+1)4^j\delta$. Their commutators with ξ and the triangular flip defects are at most $\eta = C(M+1)^3 4^M\delta$.

For completeness, an approximately invariant suffix law differs from its earliest-bit swap by at most $CM^2\eta$. Pinch ξ in the commuting R_j to ζ; each pinching costs at most η, so $\|\xi - \zeta\|_2 \le M\eta$. Intertwining a flip with one recorded digit has error $O((M+1)\eta)$ on ζ. Inserting previous joint spectral projections does not enlarge the sum of squared branch errors, since they commute with ζ. Telescoping at most M digits and returning to ξ costs $CM^2\eta$. The initial state changes by trace norm at most 2η under its flip. Purification distance now bounds the claimed total variation.

Let q_i be the probability that $R_i, \dots, R_{M-1}$ are all one, with $q_M = 1$. The swap bound gives $|2q_i - q_{i+1}| \le CM^2\eta$, hence $q_0 \le 2^{-M} + CM^2\eta$. If $a \ge 1-4^{-M}$, then $s^j(a) = 1-4^j(1-a) \ge 3/4$ for $j < M$. Thus all R_j are one on that spectral interval. This proves (4.8).

Put $G = I - A$ and $E = 1_{[0,4^{-M}]}(G)$. Since $R\xi G = RG\xi + R[\xi, G]$, its norm is at most $(1+L)\delta$. On $I-E$ the inverse of G has norm at most 4^M, while $\|R\xi E\|_2 \le L\sqrt{\mathrm{Tr}(\rho E)}$. Add the two column bounds and use (4.8). Finally choose $M = \lfloor \frac{1}{3}\log_2(1/\delta) \rfloor$. Both 2^{-M} and $4^M\delta$ are of order $\delta^{1/3}$, proving the last assertion. The relaxed polynomial factor here is harmless; only exact guard cancellation is needed in the flag proof. □

5 The finite programmed presentation

Fix a finite deterministic counter program recognizing K. Counters take nonnegative integer values and instructions increment, decrement on a nonzero branch, or test zero. Such a recognizer follows from the classical universality of counter machines [20]. One direct construction stores the two halves of a Turing tape as integer-coded stacks. Pushing multiplies a stack code by a fixed base and adds a digit; popping computes quotient and remainder by that base. Increment, decrement and zero-test loops perform these operations using finitely many scratch counters. A unary input loop writes the integer input on the simulated tape. This gives one fixed finite instruction table. Add absorbing idle and halt controls.

The variables introduced below have the domains stated explicitly. Projection inclusions $Q \le F$ mean the polynomial equation $(I-F)Q = 0$. A precision model satisfies every listed commutator and every displayed defining equation to precision δ in Definition 2.1.

5.1 Three compulsory clocks and a carry coordinate

Use three copies of (4.1), with indices a, f, t for address, fuel and time. Complete variable sets of different clocks commute. Add an address increment V and a time increment U, satisfying (4.7) for their own clocks. The address and fuel sets commute with U, and the time and fuel sets commute with V. Add a positive contraction H with

$$H = (I - P_a) + \tfrac{1}{4}P_aT_aHT_a^*, \tag{5.1}$$

$$[P_a, T_aHT_a^*] = [H, A_a] = 0, \qquad V^*A_aV = A_a - I + \tfrac{7}{4}H. \tag{5.2}$$

Include H in the address set, commuting with all fuel and time variables. The last equality will control many address translations without an exponential precision cost.

5.2 Address decoding and reset

Put $P_{a,1} = T_a P_a T_a^*$, $E_{00} = (I - P_a)(I - P_{a,1})$ and $E_1 = P_a(I - P_{a,1})$. Add a projection L and a positive contraction R with

$$L = E_1 + E_{00} T_a^2 L T_a^{-2}, \tag{5.3}$$

$$R = (I - E_{00}) + \tfrac{1}{2} E_{00} T_a^2 R T_a^{-2}. \tag{5.4}$$

The products are polynomials even before the approximate digits commute. In exact models the cylinder

$$B_e = \left(\prod_{j<e} T_a^{2j} E_{00} T_a^{-2j} \right) T_a^{2e} E_1 T_a^{-2e} \tag{5.5}$$

selects first nonzero base-four address digit one in position e. Here each base-four address digit groups two successive binary clock bits, with the first bit least significant.

Add a reset projection Z, set $S = T_f T_t$, and impose

$$Z = (I - P_f) + P_f (I - P_t) S Z S^*. \tag{5.6}$$

In exact models write $Q_j^f = T_f^j P_f T_f^{-j}$, $P_{t,j} = T_t^j P_t T_t^{-j}$ and

$$C_k = \left(\prod_{j<k} Q_j^f \right) (I - Q_k^f), \qquad D_k = \prod_{j<k} (I - P_{t,j}). \tag{5.7}$$

The product defining C_k means $(\prod_{j<k} Q_j^f)(I - Q_k^f)$, with empty product I.

5.3 Counters and transitions

Encode counter value n by $r = 2^{-n}$. For each counter add a positive contraction r_i, a zero-test projection z_i and a positive contraction s_i with

$$r_i = z_i + \tfrac{1}{2}(I - z_i) s_i. \tag{5.8}$$

All instantaneous workspace variables commute. For each control state σ, add a projection q_σ, with $q_\sigma q_\tau = 0$ for $\sigma \neq \tau$ and $\sum_\sigma q_\sigma = I$. Each program branch selector is q_σ, $q_\sigma z_i$ or $q_\sigma(I - z_i)$; these selectors partition I. The next control is the sum of selectors entering that state. The next counter is the sum, over selectors, of that selector times $r_i/2$, r_i or $2r_i$ as its instruction dictates. The factor two is used only on a nonzero decrement branch. These prescriptions define a fixed rational polynomial map Step on counters and controls.

Initialization is

$$\mathrm{Init}(q_{\mathrm{start}}) = L, \quad \mathrm{Init}(q_{\mathrm{idle}}) = I - L, \quad \mathrm{Init}(r_1) = R, \quad \mathrm{Init}(r_i) = I \ (i > 1),$$

with other controls zero. For each counter or control W, impose

$$ZW = Z\,\mathrm{Init}(W), \tag{5.9}$$

$$U^* W U = Z_+\,\mathrm{Init}(W) + (I - Z_+)\,\mathrm{Step}(W), \qquad Z_+ = U^* Z U. \tag{5.10}$$

The witnesses and zero-tests satisfy (5.8) at every time; separate update equations for them are unnecessary. Add a flag projection F and require

$$q_{\mathrm{halt}} \leq F. \tag{5.11}$$

5.4 Commutations and the conditional clock

The following is the complete additional commutation list.

- L, R, F commute with each other, with A_a, and with every fuel and time clock variable, including U.
- Z and all instantaneous workspace variables commute with one another and with A_a, A_f, A_t, L, R, F.
- The address set, including V, H, commutes with the complete fuel and time sets.

Workspace is not required to commute with U or the address shifts and flips.

For the conditional clock add positive contractions B, C, a projection Q and a self-adjoint unitary Y. Impose

$$Q \le F, \qquad [B,Q] = [B,F] = [C,Q] = [C,F] = [B,C] = 0, \tag{5.12}$$

$$VBV^* = 4B - 3Q, \qquad VCV^* = 4C - 3F, \tag{5.13}$$

$$YFY = F, \qquad YQY = F - Q, \tag{5.14}$$

$$[Y, VBV^*] = 0, \qquad [Y, C] = 0. \tag{5.15}$$

Each of B, C, Q, Y commutes with A_a and every fuel and time clock variable, including U. Define

$$F_i = V^i F V^{-i}, \qquad Q_i = V^i Q V^{-i}, \qquad Y_i = V^i Y V^{-i}. \tag{5.16}$$

The intended ternary measurement is $(I - F_i, F_i - Q_i, Q_i)$. Its inactive outcome contributes no required bit; its two active outcomes are exchanged by Y_i. The coordinate C records the flags. It is needed to control commutation of their translates, which commutation with A_a alone would not imply.

This finishes the finite presentation, denoted $\mathcal{P}_K$. Every branch and every variable used above occurs in a finite list; e, k, i in later arguments index consequences, not additional defining relations.

6 Every tracial limit pays for halting

Lemma 6.1 (Forced finite initialization)**.** *In an exact model of $\mathcal{P}_K$,*

$$B_e L = B_e, \quad B_e R = 2^{-e} B_e, \qquad C_k Z = C_k D_k. \tag{6.1}$$

The zero-test equation forces $z_i v = v$ when $r_i v = v$, and $z_i v = 0$ when $r_i v = 2^{-n} v$ for an integer $n > 0$.

Proof. By Lemma 4.1, every digit is a spectral function of its clock coordinate. The required commutations imply that L, R, Z commute with the relevant digits. Expand (5.3) and (5.4) along the e successive E_{00} factors of B_e. The nonrecursive terms vanish there. At the first E_1 factor, the nonrecursive term is I. The coefficient accumulated in the second expansion is 2^{-e}. These finite expansions prove the first two identities.

In (5.6), on C_k the first k fuel bits force the recursive branch. Each supplies the next factor $I - P_{t,j}$. At the next fuel bit the constant branch is forced. Thus $C_k Z = C_k D_k$. In particular the $k = 0$ identity forces reset on $I - P_f$, so $Z = 0$ cannot be a tracial solution.

For the zero-test, z_i commutes with s_i. On its range $r_i = I$; on its kernel $0 \le r_i \le I/2$. If $r_i v = v$, the latter component of v must vanish; if $r_i v = 2^{-n} v$ with $n > 0$, the former component must vanish. The argument applies to every vector in a given subspace, including after unitary conjugation. Although $r_i = 0$ is permitted, no finite sequence of the prescribed updates starting at an ordinary encoded integer reaches it. □

Theorem 6.2 (Flag density)**.** *Every exact tracial model of $\mathcal{P}_K$ satisfies $\tau(F) \geq \theta_K$. There is a function $\omega(\delta) \to 0$ such that every weighted precision-δ model satisfies*

$$\mathrm{Tr}(\rho F) \geq \theta_K - \omega(\delta). \tag{6.2}$$

No effective modulus is asserted.

Proof. Work first in the faithful tracial representation. The clock variables satisfy Lemma 4.1. By Proposition 4.5, U increments each finite time prefix; it commutes with fuel and address variables. Fix $e \in K$ with halting time h and choose k with $2^k > h$. The operator

$$Q_0 = B_e C_k D_k (I - F)$$

is a projection. Lemma 6.1 and (5.9) set all counters and controls on its range to the initialized values. Input counter one has value $r_1 = 2^{-e}$, while other counters encode zero.

Write $W_l = U^{-l} W U^l$ and $Z_l = U^{-l} Z U^l$. On Q_0, the first k time bits are initially zero. Increment does not return them to zero before step 2^k, so

$$Z_0 Q_0 = Q_0, \qquad Z_l Q_0 = 0 \quad (1 \leq l \leq h). \tag{6.3}$$

We prove by induction that every counter and control at time $l \leq h$ has its classical scalar right action on Q_0. The assertion holds at zero by initialization. Given these right actions, the zero-test part of Lemma 6.1, applied to the conjugated workspace, forces the correct zero-test right actions as well. Every branch selector and every update polynomial therefore has its prescribed scalar right action. If $AQ_0 = aQ_0$ and $BQ_0 = bQ_0$, then $ABQ_0 = abQ_0$; no inter-time commutation is needed.

Conjugate (5.10) by U^{-l} and multiply by Q_0 on the right. The initialized polynomials are functions of L, R and constants, so retain their scalar actions. By (6.3) the reset term vanishes and the step term yields exactly the next configuration. This proves the induction. At time h,

$$U^{-h} q_{\mathrm{halt}} U^h Q_0 = Q_0.$$

Since $q_{\mathrm{halt}} \leq F$ and $[F, U] = 0$, the left side is killed by $I - F$. Thus $Q_0 = 0$.

It remains to remove the auxiliary conditions, which must not be assumed to have uniform conditional mass. The operator $B_e(I - F)$ commutes with all fuel and time flips. Conjugate $B_e(I - F)C_k D_k = 0$ by the flip of the earliest specified fuel digit. It preserves every later specified digit. Adding the two equations removes that digit. Repeat in increasing order to remove all fuel conditions, then do the same for time. The result is $B_e(I - F) = 0$. The address cylinders B_e are orthogonal and have trace $4^{-(e+1)}$ by (4.3). Taking finite sums over halting inputs and then their supremum proves $\tau(F) \geq \theta_K$.

For the uniform approximate statement, suppose instead that a sequence of models has $\delta_j \to 0$ but $\mathrm{Tr}(\rho_j F) \leq \theta_K - \gamma$ for some $\gamma > 0$. Form their states on the universal C^*-algebra of the finitely many bounded-domain variables before imposing equations. A weak-* limit of moment states exists. It is tracial, since

$$\|[\rho_j, s]\|_1 \leq 2 \|[\sqrt{\rho_j}, s]\|_2 \longrightarrow 0$$

for each generator and hence each fixed word. Here $\|\cdot\|_1$ denotes trace norm; the inequality follows by expanding $[\xi^2, s]$ and applying Hilbert–Schmidt Cauchy–Schwarz. For every relation r, the limit has $\tau(r^* r) = 0$. Traciality gives, for bounded a,

$$\tau(a^* r^* r a) = \tau(r^* r a a^*) \leq \|a\|^2 \tau(r^* r) = 0.$$

Thus all relations are zero operators in the tracial GNS representation. This exact model has $\tau(F) \leq \theta_K - \gamma$, contradicting the first part. Taking the supremum of the positive shortfall over precision-δ models defines a function $\omega(\delta) \to 0$ and proves (6.2). □

The proof treats only finite computations before taking compactness. It never tries to quantify how far a given finite precision sees into the halting set. This compactness argument supplies no effective modulus for the uniform flag-density estimate.

7 Simultaneous finite models

Theorem 7.1. *Let $h(e)$ be the recognizer's halting time, or ∞ if it does not halt. For every even integer $N \geq 2$, define*

$$K_N = \{e < N/2 : h(e) < 2^N\}, \qquad \theta_N = \sum_{e \in K_N} 4^{-(e+1)}.$$

There is a flat model of $\mathcal{P}_K$ on dimension

$$d_N = 2^{3N+k_N}, \qquad k_N = \theta_N N + O(\log(N+2)), \tag{7.1}$$

with every defining defect at most $3 \cdot 4^{-N}$ in operator norm. The coefficients θ_N increase to θ_K.

Proof. Take N-bit address, fuel and time registers, with the finite coordinate (4.6). Their bit zero is least significant. Let the address integer be $x \in \mathbb{Z}/2^N\mathbb{Z}$ and put

$$f_N(x) = \sum_{e \in K_N} 1_{\{x \equiv 4^e \pmod{4^{e+1}}\}}. \tag{7.2}$$

The residue classes are disjoint and their periods divide 2^N, so this is a Boolean function of mean θ_N.

Every interval of N consecutive addresses has

$$\sum_{j=1}^{N} f_N(x-j) = N\theta_N + O(\log(N+2)) \tag{7.3}$$

uniformly in x, including cyclic intervals. To prove it, put $m = \lfloor \log_4 N \rfloor$. For each $e \leq m$, one residue class has a count differing from its mean by at most one. Every flagged address from $e > m$ is a multiple of 4^{m+1}, so the whole remaining family contributes at most one address to such an interval. Its expected count is at most $N/(3 \cdot 4^{m+1})$. Summing proves (7.3).

Set

$$k_N = 1 + \max_x \sum_{j=1}^{N} f_N(x-j) \tag{7.4}$$

and add k_N optional qubits. Let R_0 rotate their contents from slot i to slot $i+1$ cyclically. Define

$$V|x,y\rangle = |x+1, R_0^{f_N(x)} y\rangle, \qquad F = f_N(x), \quad Q = FP_{\text{slot }0}, \quad Y = (I-F) + FX_{\text{slot }0}. \tag{7.5}$$

Define the translates F_j, Q_j by (5.16) and set

$$B = \sum_{j \geq 0} 3 \cdot 4^{-j-1} Q_j, \qquad C = \sum_{j \geq 0} 3 \cdot 4^{-j-1} F_j. \tag{7.6}$$

These are explicit finite diagonal matrices. Indeed V is a permutation of a finite basis, so the series are periodic geometric sums. They are positive contractions. Their terms commute, so (5.12) is exact. Shifting the sums proves (5.13) exactly; the controlled flip gives (5.14) exactly.

At an active Q_j and a fixed current address x, its occupied slot is

$$\sum_{r=x-j}^{x-1} f_N(r) \pmod{k_N}.$$

If $1 \le j \le N$, this sum is positive because $f_N(x-j) = 1$, and is at most $k_N - 1$ by (7.4). Thus Y preserves Q_j for those j. It always commutes with C, and the uncontrolled remainder satisfies

$$\|[Y, VBV^*]\| \le 2 \sum_{j>N} 3 \cdot 4^{-j} = 2 \cdot 4^{-N}. \tag{7.7}$$

This verifies (5.15).

To couple V to the address clock, let T_0, X_0 be its ordinary cyclic rotation and first-bit flip and put

$$h_N(x) = \sum_{0 \le r < x} f_N(r), \quad D = \bigoplus_x R_0^{h_N(x)}, \quad T_a = D(T_0 \otimes I)D^*, \quad X_a = D(X_0 \otimes I)D^*. \tag{7.8}$$

The coordinate A_a and digit P_a commute with D, so all address clock defect bounds from Theorem 4.4 are unchanged. In the D^* gauge, V is ordinary increment except at the all-ones address, where it carries a fixed optional-register permutation. The recursive increment agrees away from that address. At all ones its two output addresses are distinct, with the same optional permutation. The guard is 4^{-N} there. Hence its operator defect is $\sqrt{2}\,4^{-N}$ and its flat defect is $\sqrt{2}\,4^{-N}2^{-N/2}$.

For $H(x)$ use 4^{-k} when the address has $k < N$ initial ones and then a zero, and use zero at all ones. Rotation removes one initial one, so (5.1) is exact, including its all-ones case. All commutations in (5.2) are exact, and off all ones

$$A_a(x+1) - A_a(x) = -1 + \tfrac{7}{4}\,4^{-k}.$$

At all ones the defect of the last equation is 4^{-N}. Its flat norm is $4^{-N}2^{-N/2}$. Let U be ordinary cyclic time increment and use ordinary clocks on fuel and time. Their defects have the same bounds as before.

It remains to verify the whole program on these same registers. For a nonzero address let e be the position of its first nonzero base-four digit. Set $R(x) = 2^{-e}$ and $L(x) = 1$ precisely when that digit is one. At the all-zero address set $R = L = 0$. Since N is even, rotation by two bits removes one whole base-four digit. Thus (5.3)–(5.4) hold exactly, including at the all-zero address.

For fuel with k initial ones then a zero, put

$$Z(q,t) = 1_{\{t \equiv 0 \pmod{2^k}\}}.$$

On all-ones fuel use $k = N$. When the first fuel bit is zero, $Z = 1$. Otherwise the factor $I - P_t$ in (5.6) requires the first time bit to be zero and the rotated equation tests the remaining $k-1$ bits. On all-ones fuel, the recursive test is true precisely at the all-zero time word. This proves (5.6) in every case.

At a basis point (x, q, t) assign every workspace counter and control the deterministic configuration

$$\mathrm{Step}^{t \bmod 2^k}(\mathrm{Init}(x)). \tag{7.9}$$

These are diagonal entries on existing registers; they add no quantum tensor factor. For ordinary counter entries set $z_i = 1, s_i = 0$ when $r_i = 1$ and $z_i = 0, s_i = 2r_i$ otherwise; the idle $r_i = 0$ case uses $z_i = s_i = 0$. These are contractions and give (5.8) exactly. The next

time point is initialized if its residue modulo 2^k is zero, and is otherwise one step of the old configuration. This is exactly (5.9)–(5.10), also at the cyclic time boundary. A displayed halt occurs in fewer than 2^N steps on an input $e < N/2$, so its address is flagged in (7.2). Hence (5.11) is exact.

All workspace and reset variables are diagonal in (x, q, t), and the optional register carries no workspace labels. Every specified remaining commutation therefore holds. The only nonzero defining defects are the clock affine and tail-flip defects, the guarded increments, the carry step and (7.7). Their operator norms are bounded by the theorem's constant. The three compulsory registers and the optional register have dimension 2^{3N+k_N}, and (7.3) gives (7.1). Finally each halting input eventually meets both cutoffs in K_N, while the address weights have a summable geometric tail. This proves $\theta_N \uparrow \theta_K$. □

8 Counting measurements on the actual state

The count must tolerate measurements that commute only approximately and a density matrix that is not flat. Its proof keeps the measurement branches together. Its last step is the usual random-access-code argument [18], using Holevo's information bound [11].

Theorem 8.1 (Triangular measurement count)**.** *Let m PVMs on $\mathbb{C}^d$ have two or three outcomes. Let ρ be a density matrix and $\xi = \sqrt{\rho}$. Assume each outcome has state commutator at most η, and outcomes of different measurements have weighted commutator at most η. For each measurement i, assume a unitary U_i has state commutator at most η, exchanges its own two binary outcomes, or its two active ternary outcomes while fixing the inactive one, and preserves every later measurement. Each of these intertwining equations is required in weighted norm with error at most η.*

If b measurements are binary and F_i is the active projection of a ternary measurement, then for an absolute constant C, with $e = Cm^4\eta^2 \le 1/2$,

$$\log d \ge b + \sum_{i \text{ ternary}} \operatorname{Tr}(\rho F_i) - Cm^3\eta - m\big[h_2(e) + e\big]. \tag{8.1}$$

Here h_2 is binary entropy, with $h_2(0) = 0$.

Proof. For a PVM (P_a) define left and right Hilbert–Schmidt isometries

$$L_P(Z) = (P_a Z)_a, \qquad R_P(Z) = (ZP_a)_a.$$

They differ on ξ by at most $\sqrt{3}\eta$. Every left multiplication commutes with every right multiplication. Moving a product of s left recording maps through ξ to right maps in reverse order therefore changes the recording vector by at most $s\sqrt{3}\eta$, by telescoping isometries.

Suppose a column of left operators $D = (D_c)_c$ has column operator norm at most L and $\|D\xi\|_2 \le \alpha$. It commutes with all right recording maps. After any prefix of s measurements, the preceding comparison gives

$$\left(\sum_{\mathbf{a},c} \|D_c P_{s,a_s} \cdots P_{1,a_1}\xi\|_2^2\right)^{1/2} \le \alpha + Ls\sqrt{3}\eta. \tag{8.2}$$

This estimates all branches together. In particular, no inverse of a small branch probability appears.

Measure in reverse index order. For the suffix starting at i, move U_i through its recorded measurements. It exchanges its own two labels and preserves later ones. A single intertwining column is a difference of two isometries, has column operator norm at most two and weighted

norm $O(\eta)$ on ξ. Equation (8.2) bounds it after any preceding measurements by $O(m\eta)$. At most m intertwiners, followed by the initial state-invariance error, give total vector error $O(m^2\eta)$. Distances of purifications bound total variation of the classical records. Thus the suffix distribution differs from its own i-th outcome swap by at most $Cm^2\eta$.

For a binary outcome with conditional law $(p, 1-p)$, concavity gives $h_2(p) \geq 1 - |2p-1|$. The suffix swap variation is exactly the average of $|2p-1|$ over the later record. Hence its conditional entropy is at least one minus that variation. For ternary probabilities,

$$H(p_0, p_1, p_2) \geq (p_1 + p_2) - |p_1 - p_2|.$$

Indeed split off the active event and apply the binary inequality within it. The ternary swap variation is the corresponding average of $|p_1 - p_2|$. The chain rule now bounds the full record entropy below by b plus recorded active probabilities minus $Cm^3\eta$.

Those recorded active probabilities can be replaced by their values on ρ. For one PVM, $(P_a\xi)_a$ and $(\xi P_a)_a$ purify, respectively, its nonselectively measured state and ρ on the original system. Their distance is at most $\sqrt{3}\eta$, so those states differ by trace norm at most $C\eta$. If Φ_j are the nonselective measurement channels, expand

$$\Phi_s \cdots \Phi_1(\rho) - \rho = \sum_{j=1}^{s} \Phi_s \cdots \Phi_{j+1}(\Phi_j(\rho) - \rho).$$

Trace-norm contractivity gives an $O(m\eta)$ bound for every prefix. Thus all active expectations change by at most $Cm^2\eta$ in total, already covered by the preceding error term.

We next bound record entropy above by dimension. Fix one recorded variable. Move its measurement to the end of the actual sequence by adjacent swaps. An elementary swap column has components $[P_a, Q_b]$, weighted norm $O(\eta)$ and column operator norm at most two: it is the difference of the two recording isometries, with record factors reordered. Equation (8.2) bounds its norm after a prefix by $O(m\eta)$. After at most m swaps the total recording vectors differ by $O(m^2\eta)$.

In the reordered process this PVM is last. Remeasuring it on the final system recovers its recorded value with zero error, since equal projective measurements are adjacent. In the original process, apply the same failure projection to the difference of recording vectors. Its squared norm bounds the decoding error by $e = Cm^4\eta^2$.

Let Z be the classical record and Q_{fin} the final system, of dimension d. The decoder just described acts only on this system, not on the purification reference. Holevo's bound, conditional entropy subadditivity and the error-indicator bound for at most three outcomes give

$$\begin{aligned} H(Z) &= I(Z : Q_{\mathrm{fin}}) + H(Z \mid Q_{\mathrm{fin}}) \\ &\leq \log d + \sum_i H(Z_i \mid Q_{\mathrm{fin}}) \\ &\leq \log d + m[h_2(e) + e]. \end{aligned}$$

For the last line, measuring the proposed decoder can only increase conditional entropy; the classical conditional entropy is at most $h_2(e) + e\log(3-1)$. Combining the upper and lower record bounds proves (8.1). □

9 The matched lower bound

Theorem 9.1. *Every weighted precision-δ model of $\mathcal{P}_K$ on $\mathbb{C}^d$ satisfies*

$$\log d \geq (3 + \theta_K)\log_4(1/\delta) - o(\log(1/\delta)). \tag{9.1}$$

The error is uniform over all states and dimensions.

9.1 The carry estimate

Identify $\mathcal{K}$ with binary sequences viewed as 2-adic integers, and let g_i add the nonnegative integer i. If two sequences first differ at digit j, their Cantor coordinates differ by between $\frac{1}{2}4^{-j}$ and 4^{-j}. Addition preserves that first differing digit. Consequently every g_i is 2-Lipschitz on $\mathcal{K}$, uniformly in i.

Let $h(a) = 4^{-k}$ if the sequence starts with exactly k ones followed by a zero, and put $h(1) = 0$. It is also 2-Lipschitz: equal first-zero positions give equal values; otherwise the smaller first-zero position j is the first differing digit, and the value difference is at most 4^{-j}. Its identities are

$$h = 1 - p + \tfrac{1}{4}p(h \circ s), \qquad g_1(a) = a - 1 + \tfrac{7}{4}h(a). \tag{9.2}$$

The latter follows by summing the coordinate change when k initial ones become zero and the next zero becomes one. The all-ones case follows by the limit $k \to \infty$. Lipschitz functions on $\mathcal{K}$ can be extended linearly across its gaps with the same constant.

Lemma 9.2 (Translations with polynomial cost)**.** *For an absolute constant C, every model of the address relations satisfies*

$$|H - h(\widehat{A}_a)|_\rho \le C\delta, \tag{9.3}$$

$$|V^{-i}\widehat{A}_a V^i - g_i(\widehat{A}_a)|_\rho \le C(i+1)\delta, \tag{9.4}$$

$$|[V^i J V^{-i}, P_{a,j}]|_\rho \le C(i+1)4^j\delta \quad (J \in \{F, Q, Y\}). \tag{9.5}$$

Proof. Let $p = p(\widehat{A}_a)$ and $E = H - h(\widehat{A}_a)$. Subtract the first identity of (9.2) from (5.1), retaining its error r. The exact subtraction is

$$\begin{aligned} E = \; & -(P_a - p) + \tfrac{1}{4}P_a T_a E T_a^* \\ & + \tfrac{1}{4}P_a(T_a h(\widehat{A}_a)T_a^* - h(s(\widehat{A}_a))) \\ & + \tfrac{1}{4}(P_a - p)h(s(\widehat{A}_a)) + r. \end{aligned}$$

The first, third, fourth and last terms have weighted size $O(\delta)$ by Lemmas 2.3 and 4.2. In the fourth term, moving the bounded right factor past ξ costs $O(\delta)$, since $h \circ s$ is Lipschitz with fixed constant. The second term is bounded by $\frac{1}{4}|E|_\rho + O(\delta)$ using $\|E\| \le 1$. Absorb it to prove (9.3). The last equation of (5.2), together with spectral repair and unitary transport, gives

$$|V^*\widehat{A}_a V - g_1(\widehat{A}_a)|_\rho \le C_1\delta.$$

Put $e_i = |V^{-i}\widehat{A}_a V^i - g_i(\widehat{A}_a)|_\rho$, so $e_0 = 0$. Since $g_i = g_{i-1} \circ g_1$ on $\mathcal{K}$ and functional calculus commutes with unitary conjugation, for $i \ge 1$

$$\begin{aligned} V^{-i}\widehat{A}_a V^i - g_i(\widehat{A}_a) = {} & V^*(V^{-(i-1)}\widehat{A}_a V^{i-1} - g_{i-1}(\widehat{A}_a))V \\ & + (g_{i-1}(V^*\widehat{A}_a V) - g_{i-1}(g_1(\widehat{A}_a))). \end{aligned}$$

The bracket in the first term is a difference of positive contractions, so it has norm at most one, and Lemma 2.2 bounds the first term by $e_{i-1} + \|[V, \xi]\|_2 \le e_{i-1} + \delta$. Both $V^*\widehat{A}_a V$ and $g_1(\widehat{A}_a)$ have spectrum in $\mathcal{K}$, where g_{i-1} is 2-Lipschitz. Hence (2.3), the bound $\|[g_1(\widehat{A}_a), \xi]\|_2 \le 2\|[\widehat{A}_a, \xi]\|_2$ and (4.4) bound the second term by $2(C_1 + 4C)\delta$. Thus $e_i \le e_{i-1} + C_2\delta$ with C_2 independent of i, and induction gives $e_i \le C_2 i\delta$. This proves (9.4). Each step pays one transport of the accumulated error and one functional-calculus loss with the uniform constant 2; no iterated factor four occurs.

Put $f_j = p \circ s^j$. Replace $P_{a,j}$ by $f_j(\widehat{A}_a)$, at cost $C4^j\delta$ by (4.5), and conjugate by V^{-i}, at transport cost $i\delta$. Then apply (2.3) to (9.4), with the Lipschitz constant $2 \cdot 4^j$ of f_j. The result is

$$|V^{-i}P_{a,j}V^i - f_j(g_i(\widehat{A}_a))|_\rho \le C(i+1)4^j\delta.$$

The composition $f_j \circ g_i$ has Lipschitz constant at most $4 \cdot 4^j$ on $\mathcal{K}$. Each $J \in \{F, Q, Y\}$ commutes with A_a to precision δ, and hence with $\widehat{A}_a$ to precision $C\delta$. Equation (2.4) bounds its commutator with this composition by $C4^j\delta$. Replace back at cost $C(i+1)4^j\delta$: for the self-adjoint difference Δ just estimated, $|[J,\Delta]|_\rho \le 2|\Delta|_\rho + \|\Delta\|\|[J,\xi]\|_2$. Conjugate by V^i, at transport cost $2i\delta$. This gives (9.5). □

9.2 Joint digits and conditional measurements

Lemma 9.3. *Fix $M \ge 1$. The $3M$ compulsory binary measurements are the raw digit PVMs $(P_{c,j}, I - P_{c,j})$, where $P_{c,j} = T_c^j P_c T_c^{-j}$ for $c \in \{a, f, t\}$ and $j < M$, with flips $X_{c,j} = T_c^j X_c T_c^{-j}$. The M conditional ternary measurements are $(I - F_i, F_i - Q_i', Q_i')$ for $i < M$, with flips Y_i, where Q_i' is the repaired data projection constructed in the proof. Index the conditional measurements first, then the address, fuel and time digits, each family in increasing digit order. All commutations between outcomes of different measurements, all flip equations required in this order, and all state commutations have error at most*

$$\eta = C(M+1)^6 4^M \delta. \tag{9.6}$$

Each Q_i' differs from Q_i by weighted size at most $C\eta$, and the conditional measurements are exact PVMs on the same state.

Proof. Two elementary rules are used throughout. First, if D is self-adjoint and Y is a contraction, then $DY\xi = D\xi Y + D[Y,\xi]$ gives

$$|[D,Y]|_\rho \le 2|D|_\rho + \|D\|\,\|[Y,\xi]\|_2. \tag{9.7}$$

Hence, if $X = X_0 + D_X$ and $Y = Y_0 + D_Y$ with self-adjoint corrections, the identity

$$[X,Y] = [D_X, Y] + [X_0, D_Y] + [X_0, Y_0]$$

shows that replacing both sides of a commutator by nearby operators adds the two replacement errors; it never multiplies them. Second, suppose every letter of a word w commutes to precision δ with every letter of a word w', adjoint letters included. Moving letters one at a time and applying Lemma 2.2 to each interchange gives $|[w,w']|_\rho \le C|w|\,|w'|(|w| + |w'| + 1)\delta$. We call this the word-interchange bound.

The required estimates are the following.

(i) *One compulsory clock.* Lemma 4.2 gives the digit commutations, own-bit flips, later-digit preservation and state commutators within each clock, with error $C(M+1)^3 4^M\delta$. Its raw digit commutators already replace both digits by commuting spectral digits, at additive cost.

(ii) *Different compulsory clocks.* For $c \ne c'$, $P_{c,j}$ and $P_{c',k}$ are words of lengths $2j+1$ and $2k+1$ in two complete clock sets, which commute by hypothesis. The word-interchange bound gives $|[P_{c,j}, P_{c',k}]|_\rho \le C(M+1)^3\delta$. The same bound shows that $X_{c,j}$ preserves every $P_{c',k}$. These estimates must go through the raw conjugated digits. Comparing the spectral digits $p(s^j(\widehat{A}_c))$ and $p(s^k(\widehat{A}_{c'}))$ by two applications of (2.4) would multiply their Lipschitz constants. That costs order $4^{j+k}\delta$, which for j, k near M exceeds (9.6) by a factor of order $4^M/(M+1)^6$ and would defeat the choice (9.8) below. If spectral

digits were measured instead, put $D_{c,j} = p(s^j(\widehat{A}_c)) - P_{c,j}$. It is self-adjoint, with norm at most one and $|D_{c,j}|_\rho \le C4^j\delta$ by (4.5). Then

$$[p(s^j(\widehat{A}_c)), p(s^k(\widehat{A}_{c'}))] = [D_{c,j}, p(s^k(\widehat{A}_{c'}))] + [P_{c,j}, D_{c',k}] + [P_{c,j}, P_{c',k}].$$

By (9.7), with $\|[p(s^k(\widehat{A}_{c'})), \xi]\|_2 \le C4^k\delta$ from Lemma 2.3 and $\|[P_{c,j}, \xi]\|_2 \le (2j+1)\delta$, its weighted norm is at most $C4^M\delta + C(M+1)^3\delta$. Thus either choice satisfies (9.6); we fix the raw digits.

(iii) *Conditional against conditional.* Use the affine-only part of the spectral repair on (B, Q, V) and (C, F, V). If $i \le j$,

$$[Q_i, F_j] = V^i[Q, F_{j-i}]V^{-i}.$$

Recover F_{j-i} from the rounded flag coordinate and apply (2.4) with $[Q, C] = 0$. The error is $O(4^{j-i}\delta)$ plus polynomial transport. If $i > j$, use $[F, B] = 0$ instead. The same argument with $[Q, B] = 0$ controls $[Q_i, Q_j]$, and with $[F, C] = 0$ controls $[F_i, F_j]$. Only one side is replaced by a coordinate function here, so there is a single depth loss.

(iv) *Conditional against address digits.* By (9.5) with $J \in \{F, Q\}$, the commutators of F_i and Q_i with $P_{a,j}$ are at most $C(M+1)4^M\delta$. With $J = Y$, the same bound shows that Y_i preserves every address digit.

(v) *Conditional against fuel and time digits.* The operators F_i, Q_i and Y_i are words of length $2i+1$ in V, V^* and one of F, Q, Y. Each of these letters commutes with every fuel and time clock variable by hypothesis. The digits $P_{f,k}$ and $P_{t,k}$ are words of length $2k+1$ in the fuel and time sets. The word-interchange bound gives commutators at most $C(M+1)^3\delta$, and shows that Y_i preserves every fuel and time digit. These bounds are again taken through the raw conjugated words. Writing F_i or Q_i as a digit of a conditional coordinate, and a fuel or time digit as a spectral digit, would stack two depth losses.

(vi) *Flips.* The own conditional flip is the V^i-translate of (5.14). For later flags, $[Y, C] = 0$ and digit recovery imply preservation by Y_i. For later data, use $[Y, VBV^*] = 0$ and digit recovery on the tail coordinate, with depth $j - i - 1$ when $j > i$. This again costs at most one factor 4^{j-i}. Conditional flips preserve all compulsory digits by (iv) and (v). A compulsory flip $X_{c,j}$ exchanges $P_{c,j}$ and preserves later digits of its own clock by (i), and every digit of a later clock family by (ii).

(vii) *State commutators.* State commutators of translated raw variables are at most $C(M+1)\delta$.

All the estimates just listed fit (9.6): cubic word-interchange costs and linear address costs are dominated by its sixth power. No factor 4^{i+j} is used.

We finally repair inclusion. For projections F, Q with $|(I-F)Q|_\rho \le \eta$ and state commutators at most η, put

$$A_0 = FQF, \quad L_0 = (I-F)QF, \quad Q' = 1_{[1/2,1]}(A_0) \le F.$$

The weighted error rule and adjoints give $|Q - A_0|_\rho \le 4\eta$ and $|L_0|_\rho \le 2\eta$. Since $A_0 - A_0^2 = L_0^*L_0$ and

$$|1_{[1/2,1]}(t) - t|^2 \le t(1-t) \quad (0 \le t \le 1),$$

we get $|Q' - A_0|_\rho \le 2\eta$ and thus $|Q' - Q|_\rho \le 6\eta$. This difference is self-adjoint, so its state commutator is at most twice its weighted norm. Replacing Q by Q' in a commutator

or a flip equation therefore costs only $C\eta$, by Lemma 2.2. Apply this independently to the raw pairs F_i, Q_i, after absorbing their polynomial transport constants into (9.6). The outcomes $(I - F_i, F_i - Q_i', Q_i')$ are now exact PVMs. All previous estimates survive with a fixed factor. □

Proof of Theorem 9.1. Index the optional measurements first, then the three compulsory clock families, in increasing digit order within each family. Optional flips preserve all compulsory measurements. A compulsory flip needs to preserve only later compulsory measurements, not the earlier optional family. Thus this ordering meets the hypotheses of Theorem 8.1, with $m = 4M$ and $b = 3M$.

Put $t = \log_4(1/\delta)$ and, for sufficiently small δ, choose

$$M = \lfloor t - 30 \log_4(2 + t) \rfloor . \tag{9.8}$$

Then $4^M \delta \le (2 + t)^{-30}$, so (9.6) gives $\eta \le C(1 + t)^{-24}$. The error terms of (8.1) are $o(1)$. Consequently

$$\log d \ge 3M + \sum_{i<M} \operatorname{Tr}(\rho F_i) - o(1).$$

Unitary state transport gives

$$|\operatorname{Tr}(\rho F_i) - \operatorname{Tr}(\rho F)| \le \|V^{-i} \rho V^i - \rho\|_1 \le 2i\delta.$$

Apply Theorem 6.2 to the same ρ. The result is

$$\log d \ge (3 + \theta_K - \omega(\delta))M - o(M).$$

Since $M = t - O(\log(t + 2))$ and $\omega(\delta) \to 0$, this is (9.1). Every estimate used the actual model state. □

Remark 9.4 (Where the guards are used). Two structural facts about the proof follow from tracing where each relation enters.

(a) The guarded relations (4.7) are used only in the exact tracial limit. In the proof of Theorem 1.1 they enter only through Proposition 4.5, in the exact part of Theorem 6.2, and the approximate part of that theorem reaches the exact case by compactness, not by a quantitative estimate. The quantitative lower bound of this section uses instead the unguarded carry equation, the last relation of (5.2), whose defect in the finite models is 4^{-N} (Theorem 7.1). Consequently Proposition 4.6 is not used in the proof of Theorem 1.1, and its guard-removal loss, of order $\delta^{1/6}$ up to logarithms, is never paid.

(b) The guarded increment (4.7) for the address increment V is used by no proof. Section 6 applies Proposition 4.5 only to U, and V acts on the address digits only through the carry relations (5.1)–(5.2), in Lemma 9.2. That relation is therefore redundant. It is harmless: the finite models satisfy it with operator defect $\sqrt{2}\, 4^{-N}$ (Theorem 7.1).

10 The game and its exponent

Corollary 10.1 (Matched source profiles)**.** *The presentation $\mathcal{P}_K$ satisfies*

$$\log W(4^{-N}) = (3 + \theta_K)N + o(N) = \log H(4^{-N}). \tag{10.1}$$

One computable comparison sequence is $A_N = 3N + \lfloor N\theta_N \rfloor$, and both logarithmic profiles equal $A_N + o(N)$.

Proof. Theorem 9.1 is the lower bound for W, and $W \le H$. Theorem 7.1 is the matching upper bound for H at a fixed multiple of the requested precision. Replacing N by an even integer at a fixed greater distance absorbs that multiple. Such a change costs $O(1)$ depth, and the upper and lower limiting slopes are both $3+\theta_K$ since $\theta_N \to \theta_K$. Monotonicity interpolates intermediate depths. Finally $N\theta_N = N\theta_K + o(N)$, giving the comparison sequence. □

The Boolean correspondence theorem [7, Theorem 4.9] supplies both game directions with the actual marginal state kept. We do not reprove that general correspondence or its return theorem. To make the precision accounting for this construction transparent, we give the elementary matrix estimate responsible for its square-root extraction and explain how it applies here.

Lemma 10.2 (Consistency and the square root of the state)**.** *Let A, B be reflections and Ξ a rectangular amplitude matrix with $\rho_B = \Xi\Xi^*$. If $\|B\Xi - \Xi A\|_2 \le \eta$, then*

$$\|[B, \sqrt{\rho_B}]\|_2 \le \sqrt{2}\,\eta. \tag{10.2}$$

Proof. Set $Y = (\Xi + B\Xi A)/2$. Then $BY = YA$, so B commutes with YY^* and its square root. Also $\|\Xi - Y\|_2 \le \eta/2$. For any rectangular matrices M, N,

$$\|\sqrt{MM^*} - \sqrt{NN^*}\|_2 \le \sqrt{2}\,\|M - N\|_2. \tag{10.3}$$

Indeed apply the eigenbasis proof of (2.2) to $f(t) = |t|$ and the self-adjoint block matrices $\left(\begin{smallmatrix} 0 & M \\ M^* & 0 \end{smallmatrix}\right)$ and its N analogue, with $Z = I$. Their absolute values are block diagonal with square-root blocks. The right-hand squared norm is $2\|M-N\|_2^2$. Discard the second left-hand block to obtain (10.3). Since B commutes with $\sqrt{YY^*}$, its commutator with $\sqrt{\rho_B}$ is bounded by twice their difference, proving (10.2). □

Proposition 10.3 (The two transfers needed here)**.** *Let G be the fixed context-variable game supplied by Theorem 3.1 for $\mathcal{P}_K$. There are constants $c, C > 0$ such that a loss-ε strategy with local dimensions at most d gives a weighted source model of precision $C\sqrt{\varepsilon}$ in dimension at most cd. A flat source model of precision δ in dimension d gives a strategy of dimension at most cd and loss at most $C\delta^2$.*

Proof. The upper direction is precisely the explicit flat strategy in Theorem 3.1. For the lower direction apply the established Boolean correspondence, followed by that theorem's weighted soundness. The correspondence is applicable because every compiled context is an ordinary Boolean predicate with a nonempty set of valid answers and every tested incidence has fixed positive probability.

Here is also a direct check of the precision ingredients in its extraction, without invoking its more general return theorem. A mixed strategy may first be replaced by a pure component of at least the same winning probability. Compressing to Schmidt supports gives local dimension at most d. Alice's context POVMs have simultaneous fixed-size ancilla dilations, and Bob's binary contractions have reflection dilations on a common extra qubit. The dimensions grow by a fixed factor; the initial ancilla state is included in the marginal. Invalid Alice answers can be reassigned to a fixed valid answer, since an invalid answer was already a certain loss. After these reductions, consistency at a tested incidence gives

$$\|(A \otimes I - I \otimes B)\psi\| \le 2\sqrt{\varepsilon/\pi_{\min}}.$$

Pull a fixed Bob context word across to Alice in reverse order. Every replacement costs this bound, and Alice's context observables commute and satisfy their truth table exactly. Thus each Bob context commutator and forbidden assignment polynomial has error $C\sqrt{\varepsilon}$ on the

marginal. Lemma 10.2 gives the same order for its state commutators, with no additional square root. These are exactly the Boolean-model conditions used by the compiler. Hence the composed source precision is $C\sqrt{\varepsilon}$ on that state. □

Proof of Theorem 1.1. Apply Proposition 10.3 and Corollary 10.1. For $\varepsilon = 2^{-n}$, precision $C\sqrt{\varepsilon}$ has

$$\log_4(1/(C\sqrt{\varepsilon})) = n/4 + O(1).$$

The weighted lower profile therefore gives

$$\log D_G(2^{-n}) \geq \tfrac{3+\theta_K}{4} n - o(n).$$

The flat models at precision 4^{-N} give loss at most $C16^{-N}$ and logarithmic dimension $(3 + \theta_K)N + o(N)$. Choosing an even $N = n/4 + O(1)$ at or above the required precision gives the matching upper bound. Fixed dimension multipliers add only $O(1)$ to the logarithm. Monotonicity also gives (1.1) for arbitrary $\varepsilon \downarrow 0$. These strategies have losses tending to zero, so $\omega_q(G) = 1$. A finite-dimensional perfect strategy would extract a source model at every positive precision in one fixed dimension, contradicting Theorem 9.1.

An oracle for K computes θ_K by finite summation and a geometric tail bound. Conversely, suppose membership below e is known and subtract those contributions. With $w_e = 4^{-(e+1)}$, the residual is at most $w_e/3$ when $e \notin K$, and at least w_e when $e \in K$. An approximation of error less than $w_e/6$, compared with threshold $2w_e/3$, distinguishes these cases. Induction computes K from a real oracle for θ_K. Thus $\theta_K \equiv_T K$, and the same holds for $\beta = (3+\theta_K)/4$. Lemma 2.4 now excludes a computable certified convergence rate. □

How this work was done

Roles. This paper is the work of its two authors together with current frontier AI models from more than one developer. The authors set the direction throughout: they chose the questions, fixed the scope and the standards of proof and review, decided which lines to pursue and which to stop, and take full responsibility for every claim. The models did most of the exploration, derivation, computation and drafting, in a sequence of short research sprints. The written construction was preceded by separate proofs of the finite clock, guarded increment, and obstructions to exact periodic verification. A refutation-first audit then expanded the weighted carry estimates, the joint measurement count, and the simultaneous finite-model bounds. Those written proofs are incorporated here.

Validation. The evidence for the results is of two kinds at this stage: complete written proofs and computations by the scripts in `anc/`, which the reader can check. The review served to find errors; it does not replace the proofs.

- *Proofs.* The new construction and its quantitative estimates are proved in the text and appendices. The Boolean correspondence is invoked from [7, Theorem 4.9], with its local-dimension precision ingredients recorded in Section 10. The conjugacy gadgets are credited in Section 3. Results used for comparison are cited.
- *Proof audit.* A written refutation-first audit read the construction and its preceding clock and guard proofs. It found no fatal step or required change to the defining relations. It expanded the polynomial translation bound, the absence of a product of digit-depth losses, the all-branch measurement estimate and the entropy decoder, and checked the finite models with a disclosed test program.

- *Computation.* The archived finite-model script builds the operators at $N = 4, 6, 8$ with a fixed one-counter recognizer that halts exactly on even inputs. It records every defining relation, required commutation, domain and state condition, and the derived decoding and reset identities. Its defect tables are in `anc/audit/recorded/`. A separate script multiplies the two conjugacy gadgets on finite matrices. No inequality in these tests was violated. Finite checks support the proofs and do not replace them; the arbitrary-state lower bound rests on the written proof.
- *Independent proof check.* A second refutation-first proof check was made by a model of another developer. It wrote its findings on the construction before reading the audit above, and then compared the two. It found no false step and no counterexample. It identified three repairs, each with a proof, and all three are applied here. The orthogonality context of the compiler now gives the projection E its own Boolean variable, defined by an exact AND context (Section 3). The list of joint relations in Lemma 9.3 now states explicitly the commutators between different compulsory clocks and between conditional and fuel or time measurements, and bounds them through the raw conjugated digits. The translation bound (9.4) is now linear in the number of translations, and the compulsory measurements are fixed explicitly. The check also recorded the two structural facts in Remark 9.4. Its two numerical scripts, for the gadgets, the selector identities and the orthogonality repair, are in `anc/crosscheck/`.
- *Independent computation.* A separately written program, built from the construction without the archived script or its output, evaluates the finite models of Section 7 at $N = 4, 6, 8$. It stores each operator as a sum of weighted basis permutations and uses its own three-counter test program. It evaluates every defining relation of Sections 4 and 5, every required commutation and domain condition, and the identities (6.1): 911, 915 and 919 relations. It computes operator norms, by exact block decomposition, as well as flat norms. Every relation stated to be exact has defect zero. The same ten relations as in the archived script have nonzero defects. The nine that do not depend on the program have the same flat ratios, and the conditional-tail ratio agrees at $N = 4, 8$, where the two test programs have the same K_N. The operator-norm defects are $3 \cdot 4^{-N}$ for the clocks, $\sqrt{2}\,4^{-N}$ for the guarded increments, 4^{-N} for the carry step and at most $0.012 \cdot 4^{-N}$ for (5.15), and the dimension is exactly 2^{3N+k_N}. Its code and recorded output are in the repository directory `verify/`. Like the archived script, it tests the finite models, not the arbitrary-state lower bound or the compiled game.

No argument here has been formally verified in Lean. A Lean formalization is planned. The proofs, computations and review are offered for checking as they stand.

Priority. For each headline result, bounded searches of arXiv and the general literature compared the statements with the closest work; the credits they found are given at the results. They found no exact preemption of the composed fixed-game exponent, which is not an exhaustive priority verdict. The search included dimension witnesses and hyperlinear profiles, noncomputable exponents and symbolic frequencies, bounded polynomial and Boolean encodings, Cantor adding machines, and computation in finite presentations. The conjugacy gadgets are credited to Fanizza and coauthors; the measurement information bound to Holevo and the random-access-code literature; the operator-assignment interpretation to the constraint-game literature; and the state-preserving Boolean correspondence to the cited companion work. The finite presentation, its simultaneous models, and the matched same-state estimates are the specific construction proved here.

A Why exact periodic verification is insufficient

This appendix records two obstructions that explain the use of approximate clocks and vanishing guard amplitudes. Neither is a general impossibility theorem for finite presentations. For the distinction between locally invariant finite marginals and globally extendible measures, see Chazottes, Gambaudo, Hochman and Ugalde [2]; the compactness argument below uses all box sizes, not extension of one finite marginal.

Proposition A.1 (Periodic extrema)**.** *Let X be a nonempty shift of finite type on $\mathbb{Z}^r$ and f a rational local observable. Put $\alpha = \min_\mu \int f\, d\mu$, where μ ranges over invariant probability measures. If averages of periodic configurations approach α, then α is computable.*

Proof. On each finite box, consider probability distributions on locally admissible patterns and impose every translation-invariance equation whose two supports stay in the box. This is a finite rational polytope. Minimize the expected value of f at the origin, obtaining a computable rational lower bound ℓ_N. Restriction from larger boxes makes these bounds monotone. A convergent subsequence of minimizers has consistent marginals on every finite support, gives an invariant measure on X, and has expectation $\lim \ell_N$. Thus $\ell_N \uparrow \alpha$. Periodic configurations and their finite fundamental domains are enumerable; local validity and their rational averages are decidable. Their successive minimum averages give upper bounds converging to their infimum. By hypothesis that infimum is α. Search until the upper and lower bounds differ by the desired error. □

Hochman–Meyerovitch's noncomputable extremal frequencies [10] therefore cannot be combined with arbitrarily good exact periodic models of those minima. In the present upper construction the workspace histories close as labels, but the complete clocks remain approximate. Proposition A.1 does not apply to those clocks.

Theorem A.2 (Extraction under a fixed rejection gap)**.** *Suppose $\mathbb{C}^d = \bigoplus_{v\in\mathcal{V}} \mathcal{H}_v$, with $M = |\mathcal{V}|$ nonzero sector projections P_v. Unitaries $U_1, \dots, U_s$ permute the sectors exactly and satisfy $\|[U_i, \xi]\|_2 \le \delta$. For forbidden sets $B_j \subseteq \mathcal{V}$, $1 \le j \le r$, suppose*

$$D_j^* D_j \ge \kappa P_{B_j}, \qquad |D_j|_\rho \le \delta, \qquad P_{B_j} = \sum_{v\in B_j} P_v,$$

where $\kappa > 0$. Let $F = \sum_v c_v P_v$, $0 \le c_v \le 1$, and $a = M\delta$. If $2(s + r/\kappa)a^2 < 1$, a transition orbit avoiding all forbidden sets has average cost at most

$$\mathrm{Tr}(\rho F) + \sqrt{2s}\, a + 2(s + r/\kappa)a^2. \tag{A.1}$$

Proof. Let $p_v = \mathrm{Tr}(\rho P_v)$ and $q_v = \sqrt{p_v}$. If π_i is the sector permutation of U_i, the reverse triangle inequality applied to the row blocks $P_v\xi$ and $P_v U_i \xi U_i^*$ gives

$$\sum_v (q_v - q_{\pi_i(v)})^2 \le \delta^2.$$

The union B of forbidden sets has $p(B) \le (r/\kappa)\delta^2$. For an orbit O, let $E_O = \sum_i \sum_{v\in O}(q_v - q_{\pi_i(v)})^2$, so $\sum_O E_O \le s\delta^2$. A simple path in the connected orbit graph has at most $|O| - 1$ edges; Cauchy–Schwarz gives

$$(q_v - q_w)^2 \le (|O| - 1)E_O.$$

Choose a bad vertex w_O in each invalid orbit. Then

$$\sum_{v\in O} q_v^2 \le 2M q_{w_O}^2 + 2M^2 E_O.$$

Summing shows that the mass b of invalid orbits is at most $2(s + r/\kappa)a^2$.

Average p uniformly within each orbit, obtaining $\bar{p}$, and put $u_v = \sqrt{\bar{p}_v}$. The variance identity and the path bound give

$$\sum_{v \in O} (q_v - \bar{q}_O)^2 \le M^2 E_O,$$

where $\bar{q}_O$ is the ordinary average of q on O. If the orbit has mass $p_O > 0$, set $t_O = (\sum_{v \in O} q_v)/\sqrt{|O| p_O}$. Then

$$\sum_{v \in O} (q_v - u_v)^2 = 2p_O(1 - t_O) \le 2p_O(1 - t_O^2) = 2 \sum_{v \in O} (q_v - \bar{q}_O)^2.$$

Consequently $\|q - u\|_2 \le \sqrt{2s}\, M\delta$. Since $\mathrm{TV}(p, \bar{p}) \le \frac{1}{2}\|q - u\|_2 \|q + u\|_2 \le \|q - u\|_2$, orbit averaging changes expected cost by at most this amount. Conditioning $\bar{p}$ on valid orbits changes total variation by $b < 1$. The result is a mixture of uniform valid-orbit laws, one of which has cost at most (A.1). □

Corollary A.3. *Assume valid finite orbits are effectively enumerable, have computable rational costs, and every such cost is at least a left-c.e. real θ. If near-minimizing models have flag cost tending to θ and $M\delta \to 0$, then θ is computable. If θ is uncomputable and flat models have flag cost tending to θ, their dimensions eventually satisfy $d \ge \kappa\delta^{-2}$.*

Proof. Theorem A.2 makes the infimum of valid finite-orbit costs equal to θ under the first hypothesis. Enumerating those costs gives computable upper bounds tending to θ; left semicomputability gives lower bounds. This computes θ.

More generally, for uncomputable θ the infimum θ_{fin} of valid finite-orbit costs must be strictly greater than θ, or there are no valid finite orbits: equality would already make θ computable by the same enumeration. If a flat model had no nonzero forbidden projection, every orbit would be valid. Exact sector permutation makes ranks constant on an orbit, so its flat cost would be a mixture of uniform orbit costs, at least θ_{fin}. Thus near-minimizing flat models eventually have a nonzero P_{B_j}. The rejection gap gives

$$\delta^2 \ge d^{-1} \operatorname{Tr}(D_j^* D_j) \ge \kappa/d.$$

This proves the claimed dimension bound. □

The semantic assumption about all valid finite-orbit costs is essential. A program orbit may fail to extend to the full clock system, or may have a cheap flag cost that the extra clock relations exclude. The corollary does not apply automatically to every operator presentation. Its flat bound explains the amplitude issue: the target dimensions here are below order δ^{-2}, so a construction satisfying these hypotheses could not use fixed-gap bad sectors. Our increment defect instead has amplitude 4^{-N} at overflow, as Proposition 4.5 proves. The initialization proof uses exact tracial cancellation and compactness, not a fixed-gap finite-sector extraction.

B Finite computations and reproducibility

The code in `anc/audit/check_models.py` represents an operator as a finite sum of weighted basis permutations. For each defining relation it enumerates all basis columns of the tensor factors that occur in that relation. The absent factors are identities, so averaging squared column norms computes exactly the full-space normalized Hilbert–Schmidt expression, before floating-point rounding. Equal output rows are combined before squaring, avoiding cancellation of large squared terms. The periodic coordinates in (7.6) are summed with integer

numerators and denominator $4^\ell - 1$ for the permutation period ℓ, then rounded once to floating point.

The test program has one counter and five controls. From the starting control, zero halts, while nonzero decrements and moves to an odd-parity control. There zero enters an absorbing nonhalting control, while nonzero decrements and returns to start. Idle and halt are absorbing. Hence input e halts exactly when it is even, at step $e + 1$. This is a test of the finite-model mechanism, not a computation of the universal halting set.

Table 1: Finite-model checks for the disclosed test program. Each ratio is a flat Hilbert–Schmidt defect divided by 4^{-N}. The first two rows apply separately to all three compulsory clocks.

Quantity	$N = 4$	$N = 6$	$N = 8$
Clock affine ratio	$3/\sqrt{2}$	$3/\sqrt{2}$	$3/\sqrt{2}$
Clock tail-flip ratio	3	3	3
Guarded increment ratio	0.353553	0.176777	0.088388
Carry-step ratio	0.25	0.125	0.0625
Conditional tail ratio	0.00585946	0.000160228	0.00256366
K_N	$\{0\}$	$\{0, 2\}$	$\{0, 2\}$
k_N	2	4	4
Dimension	16 384	4 194 304	268 435 456
Recorded checks	390	394	398

The remaining defining defects are zero to within 10^{-14} in these tests. Complete per-relation values and their reference scales are archived, not just the maxima in Table 1. The two guarded increments have identical ratios in this example. The theorem's operator-norm bounds are proved in Section 7; a script's maximum column norm is not generally an operator norm. The displayed dimensions are represented without dense matrices. The original $N = 8$ run had an observed peak working set about 1.16 GB.

The gadget script `anc/gadgets/gadget_check.py` tests self-adjointness, squares, all pairwise commutations within each context, context products, and the auxiliary conjugacies on random finite instances. It is a numerical check of the explicit extension, not a proof of the converse; that converse is proved in Section 3.

The ancillary README gives commands, dependencies, recorded outputs and limitations. All computations are run with one numerical-library thread. The finite-model script caps its exhaustive workspace implementation at $N = 8$ to stay within the stated memory budget. The independent engine described in the methods section is in the repository directory `verify/`, with its recorded output, and the two scripts of the independent proof check are in `anc/crosscheck/`.

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